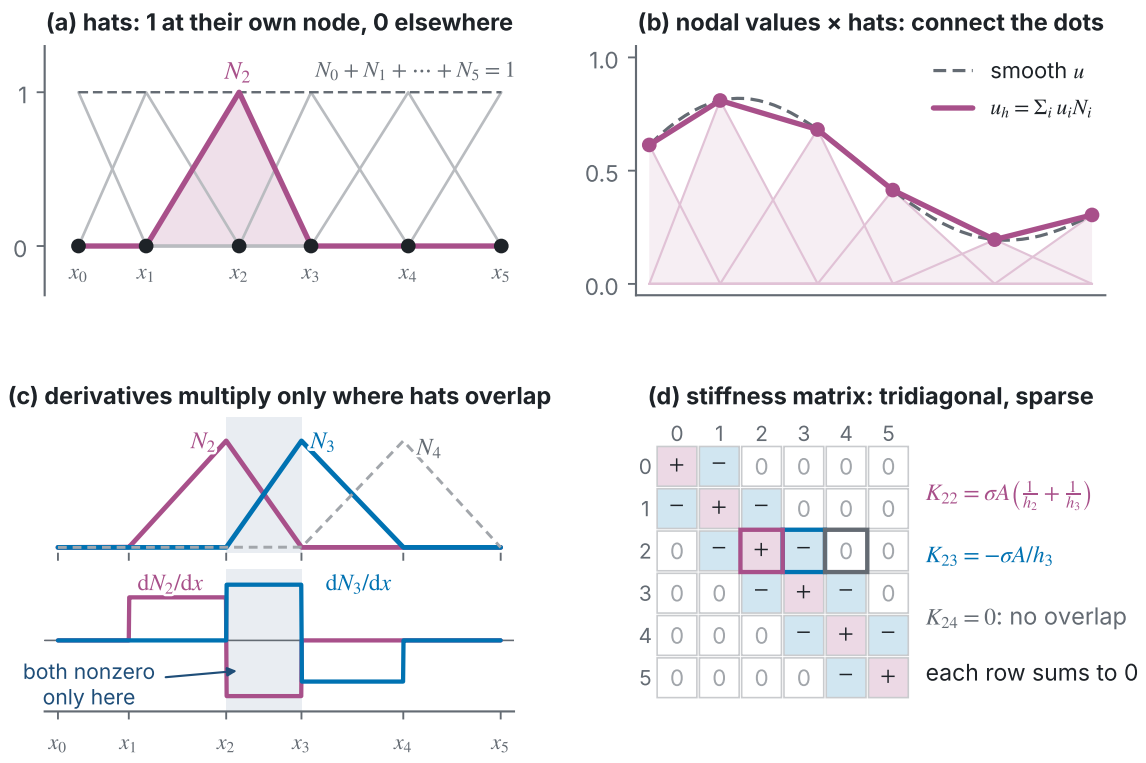

(a) hats: 1 at their own node, 0 elsewhere
N_2
$N_0 + N_1 + \cdots + N_5 = 1$
1
0
x_0 x_1 x_2 x_3 x_4 x_5
(b) nodal values × hats: connect the dots
1.0
0.5
0.0
smooth u
$u_h = \Sigma_i\, u_i N_i$
(c) derivatives multiply only where hats overlap
N_2
N_3
N_4
$\mathrm{d}N_2/\mathrm{d}x$
$\mathrm{d}N_3/\mathrm{d}x$
both nonzero
only here
x_0 x_1 x_2 x_3 x_4 x_5
(d) stiffness matrix: tridiagonal, sparse
0 1 2 3 4 5
0 + − 0 0 0 0
1 − + − 0 0 0
2 0 − + − 0 0
3 0 0 − + − 0
4 0 0 0 − + −
5 0 0 0 0 − +
$K_{22} = \sigma A\left(\frac{1}{h_2} + \frac{1}{h_3}\right)$
$K_{23} = -\sigma A/h_3$
$K_{24} = 0$: no overlap
each row sums to 0